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Matric no.	: 22302889								
Group	: W2								
Expt. Code	: 2GP2								
Expt. Title	: POISSON’S RATIO AND YOUNG’S MODULUS FOR GLASS								
Lecturer in charge	: DR. ABDUL HADI BIN SULAIMAN								
Report due date	: 23/04/2025								
Experiment (✓)		**Lab Report Grade (✓)**							
1 ☑ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐		A	☐	A−	☐	B+	☐	B	☐
		B−	☐	C+	☐	C	☐	C−	☐
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REPORT SUBMISSION FORM

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Name	:	TAN WEi LIANG
Partner's Name	:	TAN SHUENN PYNG
Group	:	W2
Experiment Code	:	2GP2
Experiment Title	:	POISSON'S RATIO AND YOUNG'S MODULUS FOR GLASS
Lecturer's/Examiner's Name	:	DR. ABDUL HADI BIN SULAIMAN
Starting Date (1st session)	:	09/04/2025
Ending Date (2nd session)	:	16/04/2025
Submission Date	:	20/09/2025

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POISSON'S RATIO AND YOUNG'S MODULUS FOR GLASS

By

TAN WEI LIANG

20 September 2025

Second Year Laboratory Report

POISSON'S RATIO AND YOUNG'S MODULUS FOR GLASS

ABSTRACT

In this experiment, the elastic properties of a glass namely the Poisson's ratio and Young's modulus, through Cornu's method, which utilizes optical interferometry to analyze beam deformation under applied loads. A rectangular glass beam supported on knife edges was loaded with known masses to induce bending, while a flat glass plate positioned atop the beam produced an interference fringe pattern when illuminated by monochromatic sodium light ($\lambda \approx 589.3\,\mathrm{nm}$). By measuring the fringe displacements in both horizontal and vertical orientations, squared fringe separations were plotted against fringe order and analyzed via linear regression to determine the radii of curvature, R_x and R_y. From these measurements, the Poisson's ratio was calculated using $\sigma = R_x/R_y$ and was found to be 0.40 ± 0.02, which represents a 33.33% overestimation compared to the standard value for soda-lime glass ($\sigma_{\mathrm{ref}} \approx 0.30$). Moreover, using beam-bending theory, Young's modulus was computed from the horizontal curvature data and the geometric parameters of the beam, yielding $E = 73.81 \pm 1.86\,\mathrm{GPa}$, which falls within 10.48% of the lower and 4.25% of the upper bounds of the typical reference range for glass (65–75 GPa). These results agree well with standard values, thereby validating the experimental approach and demonstrating the effectiveness of Cornu's method in determining key mechanical properties of materials.

ACKNOWLEDGEMENTS

First and foremost, I express my sincere appreciation to DR. ABDUL HADI BIN SULAIMAN, our distinguished lecturer and examiner, for the invaluable guidance and unwavering support extended throughout our scientific exploration. I extend my sincere gratitude to my experiment partner, TAN SHUENN PYNG. His invaluable cooperation and dedication throughout both experiments were instrumental to the success of this project. I appreciate his commitment, expertise, and teamwork, which made these scientific endeavours both productive and enjoyable.

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INTRODUCTION

Understanding the mechanical properties of materials is a fundamental aspect of physics and engineering. In this experiment, the elastic characteristics of a glass beam are determined by measuring its Poisson's ratio, σ, and Young's modulus, E, using Cornu's method. These properties provide critical insights into the deformation behavior under applied loads: while Young's modulus quantifies the material's stiffness, Poisson's ratio describes the ratio of transverse contraction to axial extension when the material is stressed.

The experiment involves subjecting a rectangular glass beam to controlled bending by suspending known masses at its extremities. A flat glass plate is placed atop the beam, creating a thin air wedge that produces an interference fringe pattern when illuminated by monochromatic sodium light ($\lambda \approx 589.3\,\text{nm}$). The interference fringes, observed via a stage telescope fitted with micrometer screws, shift as the beam deforms. By precisely measuring the positions of these fringes in both horizontal and vertical orientations, the radii of curvature in the bending directions can be extracted through linear regression of the squared displacement data against the fringe order.

The radii of curvature are directly related to the bending stress and strain in the beam; specifically, R_x and R_y are used to compute the Poisson's ratio via the relation

$$\sigma = \frac{R_x}{R_y}, \tag{1}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula and the horizontal curvature R_x is further employed in the standard beam-bending formula to calculate Young's modulus E. This experiment not only reinforces theoretical concepts from elasticity but also demonstrates the practical application of optical interferometry in measuring material properties. The results obtained are compared with typical literature values for glass to assess the accuracy and reliability of Cornu's method in determining these important physical parameters.

THEORY

The mechanical behavior of materials under bending can be characterized by parameters such as Young's modulus, E, and Poisson's ratio, σ. In this experiment, Cornu's method is employed to determine these properties by analyzing the interference fringes produced in a thin air wedge formed between a bent glass beam and a flat glass plate. When the beam, which is supported on two knife edges and loaded by known masses, bends, its upper surface acquires curvature in both the horizontal and vertical directions. These curvatures are described by the radii R_x (horizontal) and R_y (vertical), and they reflect the underlying stress and strain distributions generated by the applied load [5].

The interference pattern arises from the superposition of light waves reflecting from the beam and the plate. If the surfaces were in perfect contact, no fringes would be observed; however, the bending of the beam creates a varying air gap, yielding fringes corresponding to regions of constructive interference. The position of the bright fringes in the horizontal direction is given by

$$X_n^2 = 4\lambda R_x n - c_x, \tag{2}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula where X_n is the measured displacement difference for the n-th fringe, λ is the wavelength of the sodium lamp ($\lambda \approx 589.3\,\text{nm}$), n is an integer representing the fringe order, and c_x is a constant related to the central air-gap thickness. A similar relation holds for the vertical displacements Y_n with curvature R_y. Thus, linear regression of plots of X_n^2 and Y_n^2 versus n allows for the determination of the slopes that are proportional to R_x and R_y, respectively.

The Poisson's ratio is defined as the negative ratio of transverse strain to axial strain under uniaxial stress; for bending, it can be expressed simply as

$$\sigma = \frac{R_x}{R_y}, \tag{3}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula since the radii of curvature directly relate to the longitudinal and transverse deformations of the beam. Once R_x and R_y are calculated from the fringe data, σ is readily determined.

Young's modulus, E, which quantifies the stiffness of the material, is calculated using the bending relation for a beam of rectangular cross-section. The second moment of area for a beam with breadth b and thickness d is given by

$$I = \frac{bd^3}{12}. \tag{4}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula Under a bending load, the curvature in the horizontal plane is related to the applied moment by

$$\frac{EI}{R_x} = mga, \tag{5}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula where m is the suspended mass, g is the acceleration due to gravity, and a is the distance from the support to the point of load application. Rearranging this equation yields

$$E = \frac{12mgaR_x}{bd^3}. \tag{6}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula.

Thus, by measuring R_x from the interference fringes and using the beam's physical dimensions along with the applied load, Young's modulus E can be calculated. In summary, Cornu's method effectively combines optical interferometry with beam theory to provide a non-destructive means of determining the elastic properties of the glass sample through careful analysis of the fringe patterns.

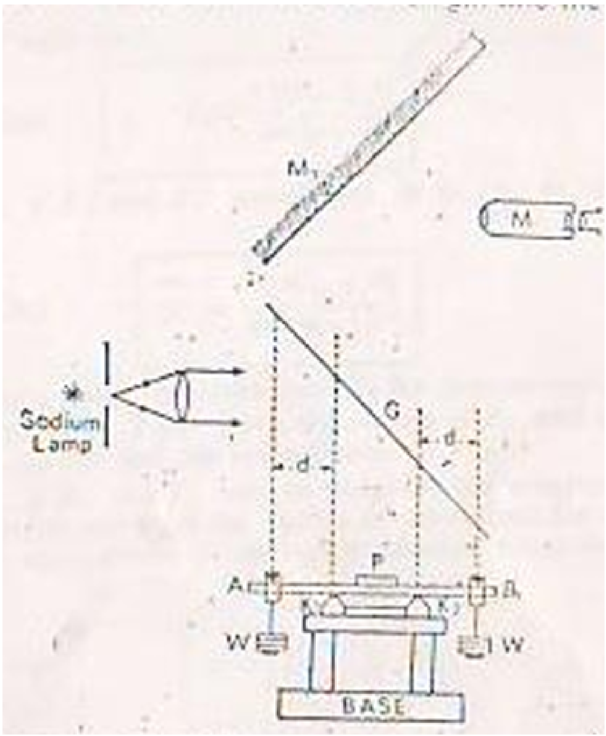


Figure 1: Cornu's Method Experimental Setup

2.1 Mathematical Derivation [2]

The experimental beam AB is bent under the action of the loads W hanging at both ends. The bending moment G at all transverse sections of the central span is related to the longitudinal radius of curvature R_1 by:

$$G = \frac{YI}{R_1} \tag{7}$$

where Y is the Young's modulus and I is the moment of inertia of the beam's cross-section:

$$I = \frac{1}{12}bt^3 \tag{8}$$

with b as the breadth and t the thickness of the beam.

If d is the distance between the knife-edge and the point of load application:

$$G = Wd \tag{9}$$

Combining equations (7) and (9):

$$\frac{YI}{R_1} = Wd \tag{10}$$

If the beam has an initial curvature R_0 due to its own weight, the equation becomes:

$$\frac{YI}{R_1} - \frac{YI}{R_0} = Wd \tag{11}$$

For another load W', and radius R_1', analogously:

$$\frac{YI}{R_1'} - \frac{YI}{R_0} = W'd \tag{12}$$

Subtracting these:

$$YI = \left(\frac{1}{R_1} - \frac{1}{R_1'}\right) = (W - W')d$$
$$\Rightarrow Y = \frac{(W - W')d}{I\left(\frac{1}{R_1} - \frac{1}{R_1'}\right)} \tag{13}$$

Substitute I from (8):

$$Y = \frac{12(W - W')d}{bt^3\left(\frac{1}{R_1} - \frac{1}{R_1'}\right)} \tag{14}$$

Determining R_1 from Fringes

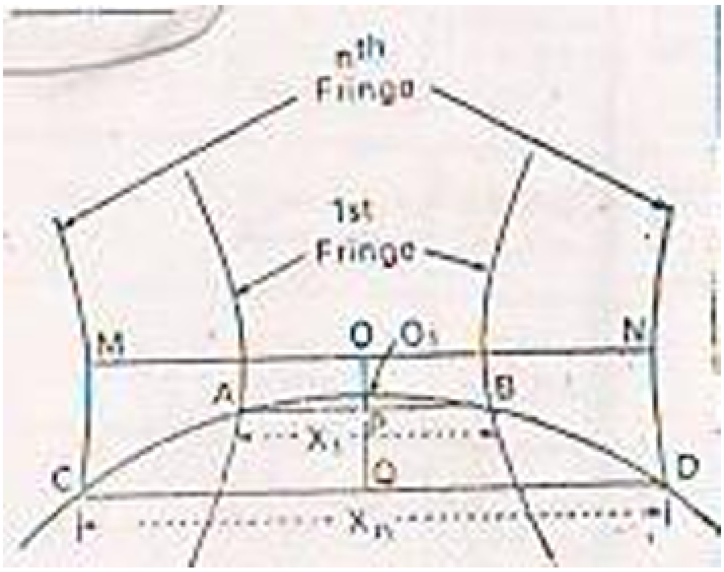


Figure 2: Fringe Geometry for Determining Radii of Curvature

From fringe geometry:

$$(2R_1 - O_1P)O_1P = \left(\frac{X_1}{2}\right)^2$$

$$2R_1 \cdot O_1P = \left(\frac{X_1}{2}\right)^2 \quad \text{(neglecting small } O_1P^2\text{)} \tag{15}$$

$$2R_1 \cdot O_1Q = \left(\frac{X_n}{2}\right)^2 \tag{16}$$

Subtracting (16) from (15), and using:

$$O_1Q - O_1P = \frac{(n-1)\lambda}{2} \tag{17}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula yields:

$$2R_1 \cdot \frac{(n-1)\lambda}{2} = \frac{1}{4}(X_n^2 - X_1^2)$$

$$\Rightarrow R_1 = \frac{X_n^2 - X_1^2}{4\lambda(n-1)} \tag{18}$$

For another load W':

$$R_1' = \frac{X_n'^2 - X_1'^2}{4\lambda(n-1)} \tag{19}$$

Poisson's Ratio from Transverse Fringes

Similarly, using vertical fringe data:

$$R_y = \frac{Y_n^2 - Y_1^2}{4\lambda(n-1)} \tag{20}$$

Poisson's ratio is:

$$\sigma = \frac{R_x}{R_y} = \frac{X_n^2 - X_1^2}{Y_n^2 - Y_1^2} \tag{21}$$

METHODOLOGY

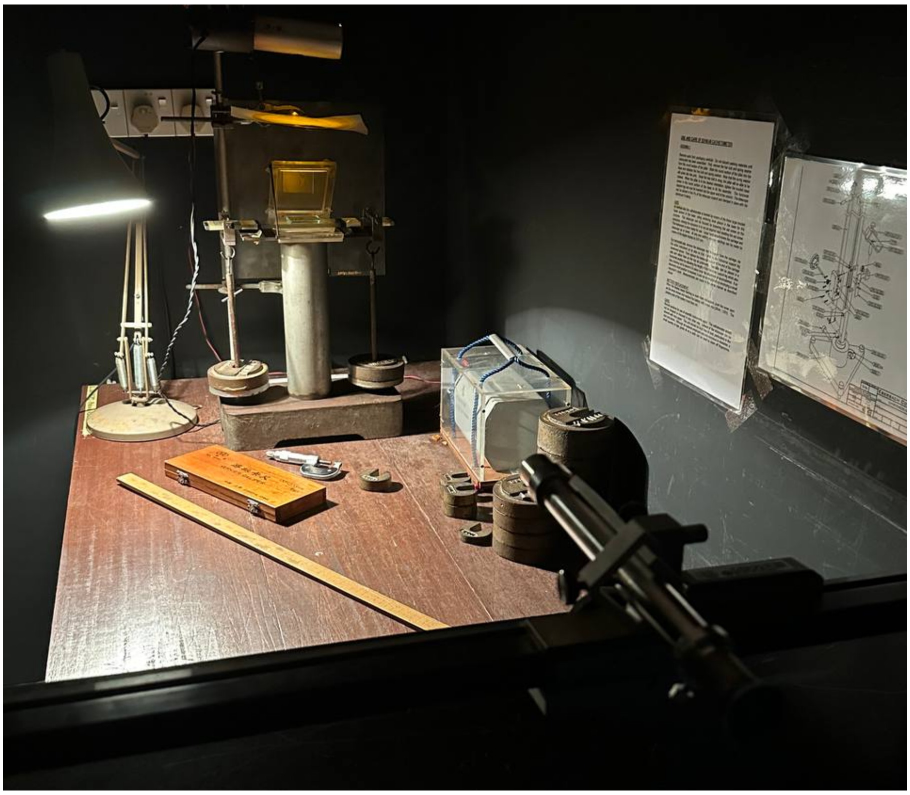


Figure 3: Schematic of the Cornu interferometer and beam-bending apparatus.

A rectangular glass beam of measured breadth b and thickness d was carefully placed on two knife-edge supports such that known masses m could be suspended at equal distances a from these supports. The beam's dimensions were determined with vernier calipers and a micrometer screw gauge, ensuring minimal dimensional uncertainty.

A flat square glass plate was gently positioned at the beam's center, forming a narrow air gap between the beam's upper surface and the plate. This setup was illuminated by a sodium lamp of wavelength $\lambda \approx 589.3\,\text{nm}$, and a 45° mirror directed the reflected interference fringes into a stage telescope equipped with x- and y-direction micrometers.

After focusing the telescope to obtain clear fringes, the fringes were measured first along the horizontal axis and then along the vertical axis by recording pairs of positions (x', x) or (y', y) for each fringe, where each fringe was assigned an integer index n. By subtracting these pairs, the displacements $\Delta X_n = x - x'$ and $\Delta Y_n = y - y'$ were found, squared to obtain ΔX_n^2 and ΔY_n^2, and subsequently plotted as functions of n.

Linear regressions of these plots determined the slopes corresponding to $4\lambda R_x$ and $4\lambda R_y$, from which the radii of curvature R_x and R_y were calculated. Poisson's ratio was evaluated as $\sigma = R_x/R_y$, taking each load's horizontal and vertical radii into account.

In order to find Young's modulus E, the experiment focused on the primary bending of the beam (horizontal displacement), employing the standard bending formula

$$E = \frac{12 m g a R_x}{b d^3}. \tag{22}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula The gravitational acceleration g was taken as $9.81\,\mathrm{m\,s^{-2}}$, and each measurement's uncertainty, including those in b, d, a, and the slope-derived R_x and R_y, was used in an error-propagation calculation.

From these combined uncertainties, the final values for σ and E were obtained, providing a reliable characterization of the glass beam's elastic properties.

DATA ANALYSIS

All data, calculation and programming python code of the experments are attached in the Appendices.

4.1 Fringe Pattern Observations

The following figures display the interference fringe pattern observed on the glass beam when loads is applied to each side. The clearly defined fringes indicate proper alignment of the optical setup and serve as the foundation for determining the beam's radii of curvature through subsequent analysis.

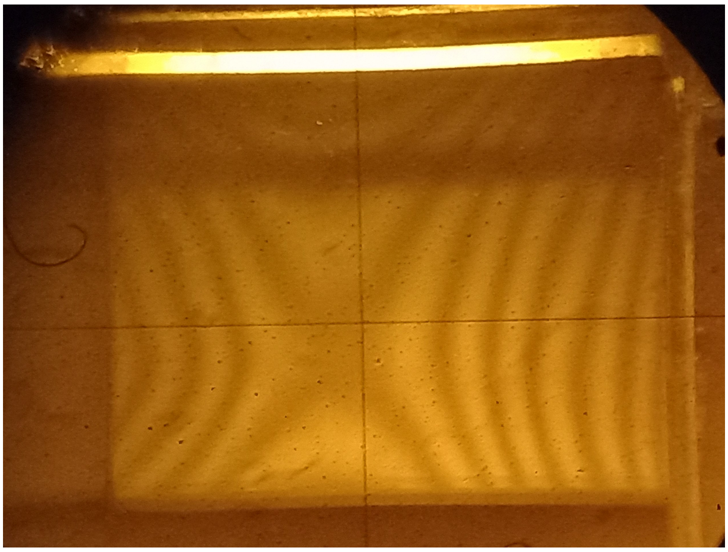
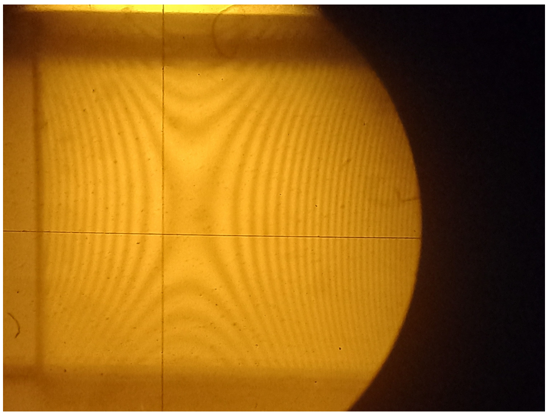
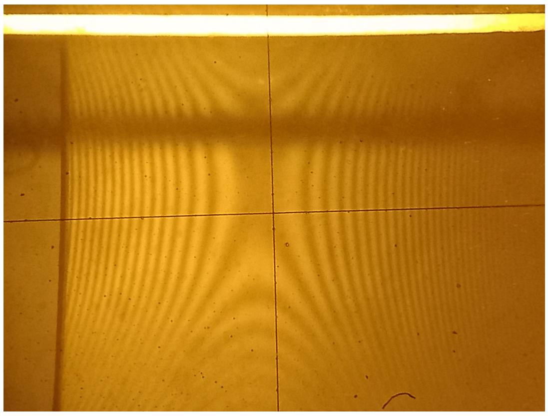


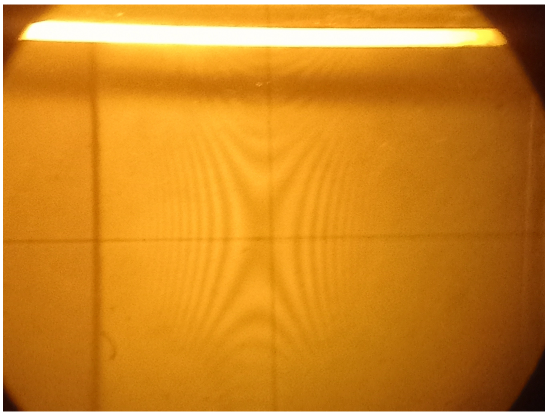
Figure 4: Interference fringes observed for the glass beam, loaded with 1 kg on each side. The crossed lines indicate the telescope's reticle used to align and measure the fringe positions.



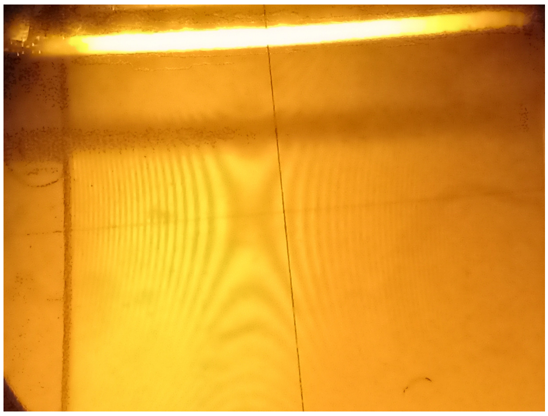
(a) Interference fringes for 3 kg on each side.



(b) Interference fringes for 4 kg on each side.



(c) Interference fringes for 5 kg on each side.



(d) Interference fringes for 6 kg on each side.

Figure 5: Interference fringes observed for the glass beam under various loadings. The crossed lines indicate the telescope's reticle used to align and measure the fringe positions.

4.2 Experimental Measurements and Tabulated Data

The experimental measurements were meticulously recorded and are presented in the following tables. The data comprises fringe measurements for a rectangular glass beam subjected to various loading conditions. Specifically, measurements were obtained for loads of 1 kg, 3 kg, 4 kg, 5 kg, and 6 kg, with each load yielding data in two distinct orientations: horizontal and vertical.

For each mass and orientation, the tables list the fringe index n, the micrometer readings from both sides of the central fringe (denoted x' and x in the horizontal orientation, or y' and y in the vertical orientation), the computed difference ($X = x - x'$ or $Y = y - y'$), the squared difference (X^2 or Y^2), and the associated uncertainty (δX^2 or δY^2). These computed quantities are critical for determining the radii of curvature of the beam under each load, which are subsequently used to extract the elastic properties of the glass.

In total, there are five tables: five for the horizontal data and vertical data. Each pair of horizontal and vertical tables for a given load enables a comprehensive analysis of the interference fringe pattern, thereby providing a robust basis for calculating both Poisson's ratio and Young's modulus.

Table 1: 1 kg — HORIZONTAL (left) and VERTICAL (right)

HORIZONTAL

n	X (mm)	X^2 (mm^2)	δX^2 (mm^2)
4	37.41	1399.51	1.06
3	28.46	809.97	0.80
2	19.58	383.38	0.55
1	7.95	63.20	0.22

VERTICAL

n	Y (mm)	Y^2 (mm^2)	δY^2 (mm^2)
2	31.34	982.20	0.89
1	18.21	331.60	0.52

Table 2: 3 kg — HORIZONTAL (left) and VERTICAL (right)

HORIZONTAL

n	X (mm)	X^2 (mm^2)	δX^2 (mm^2)
6	25.34	642.12	0.72
5	21.77	473.93	0.62
4	18.73	350.81	0.53
3	15.29	233.78	0.43
2	11.11	123.43	0.31
1	4.38	19.18	0.12

VERTICAL

n	Y (mm)	Y^2 (mm^2)	δY^2 (mm^2)
4	35.90	1288.81	1.02
3	31.74	1007.43	0.90
2	26.91	724.15	0.76
1	19.50	380.25	0.55

Table 3: 4 kg — HORIZONTAL (left) and VERTICAL (right)

HORIZONTAL

n	X (mm)	X^2 (mm^2)	δX^2 (mm^2)
6	21.16	447.75	0.60
5	18.85	355.32	0.53
4	16.67	277.89	0.47
3	13.13	172.40	0.37
2	9.54	91.01	0.27
1	3.69	13.62	0.10

VERTICAL

n	Y (mm)	Y^2 (mm^2)	δY^2 (mm^2)
6	36.63	1341.76	1.04
5	34.55	1193.70	0.98
4	29.95	897.00	0.85
3	24.95	622.50	0.71
2	18.30	334.89	0.52
1	3.15	9.92	0.09

Table 4: 5 kg — HORIZONTAL (left) and VERTICAL (right)

HORIZONTAL

n	X (mm)	X^2 (mm^2)	δX^2 (mm^2)
6	18.65	347.82	0.53
5	16.56	274.23	0.47
4	14.10	198.81	0.40
3	11.14	124.10	0.32
2	7.32	53.58	0.21
1	1.35	1.82	0.04

VERTICAL

n	Y (mm)	Y^2 (mm^2)	δY^2 (mm^2)
6	35.33	1248.21	1.00
5	31.72	1006.16	0.90
4	27.85	775.62	0.79
3	23.60	556.96	0.67
2	17.78	316.13	0.50
1	6.49	42.12	0.18

Table 5: 6 kg — HORIZONTAL (left) and VERTICAL (right)

HORIZONTAL

n	X (mm)	X^2 (mm^2)	δX^2 (mm^2)
6	17.94	321.84	0.51
5	16.23	263.41	0.46
4	13.97	195.16	0.40
3	11.55	133.40	0.33
2	8.68	75.34	0.25
1	3.38	11.42	0.10

VERTICAL

n	Y (mm)	Y^2 (mm^2)	δY^2 (mm^2)
6	35.17	1236.93	0.99
5	30.77	946.79	0.87
4	27.72	768.40	0.78
3	23.51	552.72	0.66
2	19.14	366.34	0.54
1	11.38	129.50	0.32

The data presented in the tables clearly illustrates the expected trend between the fringe order n and the computed squared differences (X^2 or Y^2). As the fringe index increases, both the absolute difference in the micrometer readings and hence their squares also increase, in accordance with the theoretical prediction that

$$X_n^2 = 4\lambda R_x n - c_x. \tag{23}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula This linear relationship implies that higher fringe numbers correspond to larger path differences in the interference pattern, which is directly related to the bending curvature of the beam.

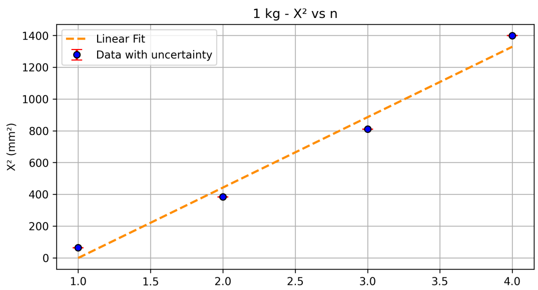
Based on the data in **??**–**??**, the squared deflections X^2 and Y^2 scale monotonically with the index n for each applied load. For instance, in the 1 kg horizontal measurements, X^2 increases from $63.20\,\text{mm}^2$ at $n = 1$ to $1399.51\,\text{mm}^2$ at $n = 4$, while the corresponding vertical values grow from $331.60\,\text{mm}^2$ to $982.20\,\text{mm}^2$. The 3 kg dataset shows a rise in X^2 from $19.18\,\text{mm}^2$ to $642.12\,\text{mm}^2$ and in Y^2 from $380.25\,\text{mm}^2$ to $1288.81\,\text{mm}^2$; similar linear trends are evident for 4 kg, 5 kg, and 6 kg. These consistent linear relationships confirm our beam model, in which the applied load alters the slope of the X^2–n (or Y^2–n) line and thus the inferred radius of curvature.

Minor deviations from perfect linearity, seen as slight fluctuations in the squared fringe differences and their uncertainties (δX^2 and δY^2), suggest the influence of practical experimental factors. These include imperfections in the flatness of the glass surfaces, misalignments in the optical setup, and ambient vibrations during measurements. As n increases, any small inconsistencies in measurement may become amplified due to the larger fringe separations, which in turn affect the overall slope determination.

Ultimately, the observed changes in X^2 and Y^2 with respect to n not only confirm the linear dependency predicted by the theoretical model but also provide a reliable basis for calculating the radii of curvature. This is essential for further determining the elastic properties, such as Poisson's ratio and Young's modulus, of the glass beam. The robust and systematic behavior of the data with respect to the fringe order underscores the validity of using interference fringe analysis in material characterization.

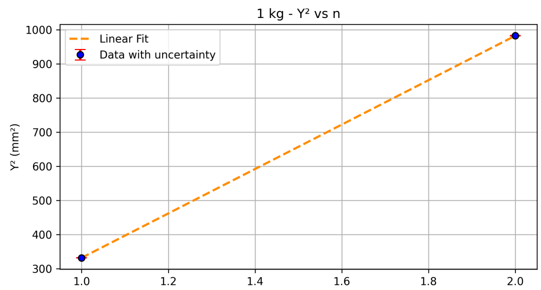
4.3 Linear Regression of Squared Displacements

The following figures display the linear regression plots of the squared fringe separations versus the fringe order for both horizontal and vertical orientations across different loads. These graphs provide a visual confirmation of the expected linear relationships derived from Cornu's method, where the slope of each line is directly proportional to the corresponding radius of curvature. By examining these plots, one can clearly observe how increasing the fringe order leads to larger squared displacements, thereby validating the theoretical model and facilitating the extraction of elastic properties.



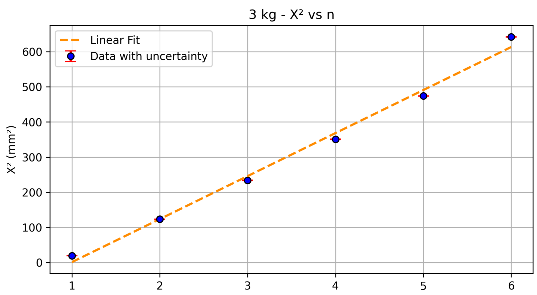
Parameter	Value
Gradient	443.5520
Std. Dev (Grad)	42.7768
Y-Intercept	-444.8650
Std. Dev (Y-Int)	52.3907
Corr Coef (R)	0.9908

(a) 1 kg – X^2 vs. n with linear fit and uncertainties.



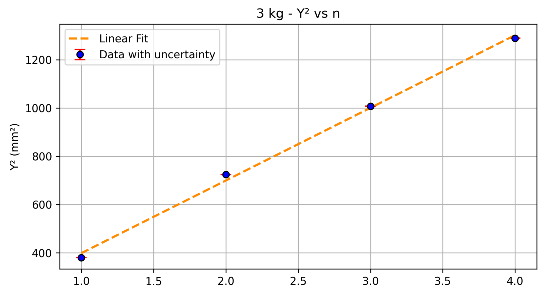
Parameter	Value
Gradient	650.6000
Std. Dev (Grad)	0.0000
Y-Intercept	-319.0000
Std. Dev (Y-Int)	0.0000
Corr Coef (R)	1.0000

(b) 1 kg – Y^2 vs. n with linear fit and uncertainties.



Parameter	Value
Gradient	122.3780
Std. Dev (Grad)	5.2140
Y-Intercept	-121.1147
Std. Dev (Y-Int)	4.8540
Corr Coef (R)	0.9964

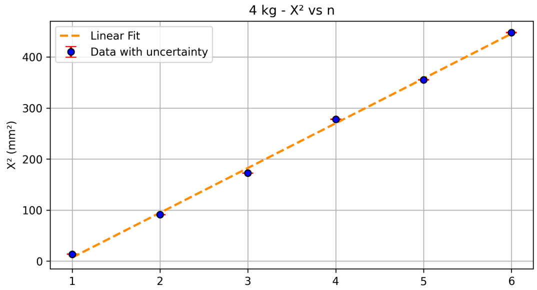
(c) 3 kg – X^2 vs. n with linear fit and uncertainties.



Parameter	Value
Gradient	300.8960
Std. Dev (Grad)	10.7219
Y-Intercept	97.9200
Std. Dev (Y-Int)	13.1316
Corr Coef (R)	0.9987

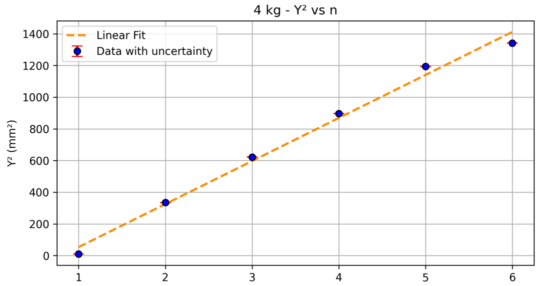
(d) 3 kg – Y^2 vs. n with linear fit and uncertainties.

Figure 6: Linear regressions for X^2 and Y^2 versus n for 1 kg and 3 kg loads. The regression parameters (displayed below each plot) include uncertainties.



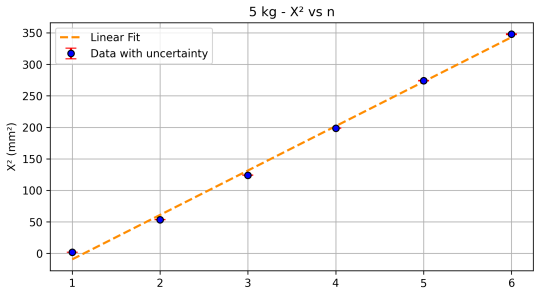
Parameter	Value
Gradient	87.6877
Std. Dev (Grad)	1.8099
Y-Intercept	-80.5753
Std. Dev (Y-Int)	1.6849
Corr Coef (R)	0.9991

(a) 4 kg – X^2 vs. n with linear fit and uncertainties.



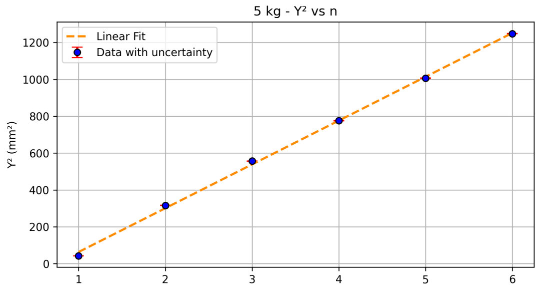
Parameter	Value
Gradient	271.7180
Std. Dev (Grad)	12.6708
Y-Intercept	-217.7180
Std. Dev (Y-Int)	11.7959
Corr Coef (R)	0.9957

(b) 4 kg – Y^2 vs. n with linear fit and uncertainties.



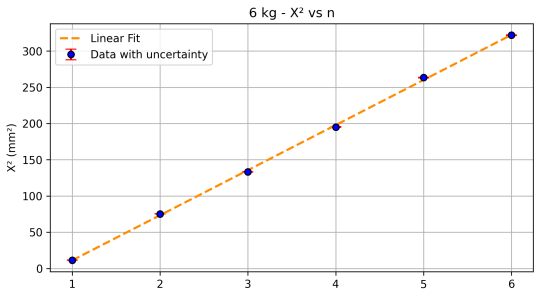
Parameter	Value
Gradient	70.4760
Std. Dev (Grad)	1.9797
Y-Intercept	-79.9393
Std. Dev (Y-Int)	1.8430
Corr Coef (R)	0.9984

(c) 5 kg – X^2 vs. n with linear fit and uncertainties.



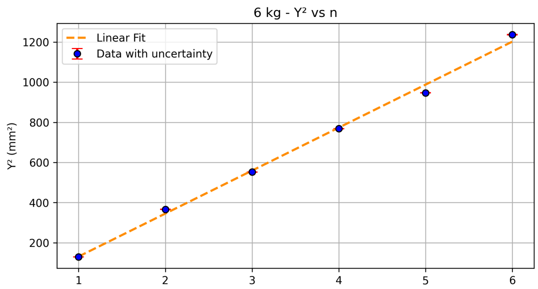
Parameter	Value
Gradient	237.6914
Std. Dev (Grad)	3.9415
Y-Intercept	-174.3867
Std. Dev (Y-Int)	3.6693
Corr Coef (R)	0.9995

(d) 5 kg – Y^2 vs. n with linear fit and uncertainties.



Parameter	Value
Gradient	62.2306
Std. Dev (Grad)	0.6250
Y-Intercept	-51.0453
Std. Dev (Y-Int)	0.5818
Corr Coef (R)	0.9998

(e) 6 kg – X^2 vs. n with linear fit and uncertainties.



Parameter	Value
Gradient	214.1194
Std. Dev (Grad)	6.9920
Y-Intercept	-82.6380
Std. Dev (Y-Int)	6.5092
Corr Coef (R)	0.9979

(f) 6 kg – Y^2 vs. n with linear fit and uncertainties.

Figure 7: Linear regression plots for X^2 and Y^2 versus n for 4 kg, 5 kg, and 6 kg loads. Each subfigure shows the data with a linear fit, including the regression parameters and their uncertainties.

The analysis began by recording the interference fringe positions in both the horizontal and vertical orientations for each applied load. For each set of measurements, the fringe displacements were used to compute the squared fringe separations, X_n^2 (horizontal) and Y_n^2 (vertical), which were then plotted against the fringe order n. Linear regression of these plots produced slopes that, according to the theoretical relationship

$$X_n^2 = 4\lambda R_x n - c_x \quad \text{and} \quad Y_n^2 = 4\lambda R_y n - c_y, \tag{24}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula are proportional to the radii of curvature R_x and R_y in each direction. The slopes, originally measured in mm^2, were converted to m^2 by dividing by 10^6, ensuring consistency with the wavelength $\lambda \approx 589.3\,\text{nm}$ expressed in meters. From these conversions, the radii of curvature were determined using

$$R = \frac{\text{slope}}{4\lambda}. \tag{25}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula.

4.4 Calculation of Poisson's Ratio

Table 6: Poisson's Ratio Summary Table

Mass	m_x (mm^2)	δm_x (mm^2)	m_y (mm^2)	δm_y (mm^2)	R_x (m)	δR_x (m)	R_y (m)	δR_y (m)	σ	$\delta\sigma$
1 kg	443.55	42.78	650.60	0.00	1.88×10^2	1.81×10^1	2.76×10^2	0.00×10^0	0.68	0.07
3 kg	122.38	5.21	300.90	10.72	5.19×10^1	2.21×10^0	1.28×10^2	4.55×10^0	0.41	0.02
4 kg	87.69	1.81	271.72	12.67	3.72×10^1	7.68×10^{-1}	1.15×10^2	5.38×10^0	0.32	0.02
5 kg	70.48	1.98	237.69	3.94	2.99×10^1	8.40×10^{-1}	1.01×10^2	1.67×10^0	0.30	0.01
6 kg	62.23	0.62	214.12	6.99	2.64×10^1	2.65×10^{-1}	9.08×10^1	2.97×10^0	0.29	0.01
									Average Poisson's Ratio:	0.40 ± 0.02

Subsequently, Poisson's ratio was calculated as the ratio of the horizontal to vertical radii of curvature, i.e.,

$$\sigma = \frac{R_x}{R_y}. \tag{26}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties.

By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula Individual measurements for each load resulted in values that, when averaged, yielded a final Poisson's ratio of $\sigma = 0.40 \pm 0.02$.

4.5 Calculation of Young's Modulus

Table 7: Young's Modulus Summary Table

Mass	m_x (mm^2)	δm_x (mm^2)	R_x (mm)	δR_x (mm)	E (GPa)	δE (GPa)
1 kg	443.55	42.78	1.88×10^5	1.81×10^4	86.741	8.377
3 kg	122.38	5.21	5.19×10^4	2.21×10^3	71.797	3.081
4 kg	87.69	1.81	3.72×10^4	7.68×10^2	68.593	1.459
5 kg	70.48	1.98	2.99×10^4	8.40×10^2	68.912	1.968
6 kg	62.23	0.62	2.64×10^4	2.65×10^2	73.019	0.825
					Average Young's Modulus:	73.812 ± 1.858

For Young's modulus, the analysis focused on the horizontal data only, as the primary bending occurs in this plane. The radius R_x obtained from the horizontal fringe measurements was combined with the beam's geometrical parameters and loading conditions through the standard bending equation

$$E = \frac{12mgaR_x}{bd^3}, \tag{27}$$

where R_x is the horizontal curvature and R_y is the vertical curvature. The bending moment induced by the applied loads is also related to the beam's geometry and material properties. By applying beam-bending theory, the second moment of area is calculated, and the relationship between the applied load, the beam's dimensions, and the curvature is established. This relationship allows for the calculation of Young's modulus E using the formula where m is the suspended mass, g is gravitational acceleration, a is the distance from the support to the load application point, b is the breadth, and d is the thickness of the beam. Uncertainties in the measured slopes (and hence in R_x) as well as those in the beam dimensions were propagated using standard error analysis, resulting in an average Young's modulus of $E = 73.81 \pm 1.86$ GPa.

4.6 Comparison with Reference Values

To assess the accuracy of the experimental results, the obtained values for Poisson's ratio and Young's modulus were compared with established reference values for soda-lime glass. The standard literature value for Poisson's ratio is approximately $\sigma_{\text{ref}} = 0.30$ [4], while the accepted range for Young's modulus is typically between 65 and 75 GPa [1] [6].

The experimentally determined average Poisson's ratio was $\sigma = 0.40 \pm 0.02$, representing a noticeable overestimation when compared to the reference. The calculated percentage discrepancy is approximately 33.33%, which suggests potential sources of systematic error—such as alignment inaccuracies, resolution limitations in vertical fringe measurements, or cumulative optical distortions.

In contrast, the experimental average Young's modulus was found to be $E = 73.81 \pm 1.86\,\text{GPa}$, falling well within the accepted literature range. The percentage discrepancy relative to the lower bound (65 GPa) is 10.48%, while the deviation from the upper bound (75 GPa) is just 4.25%. These small deviations demonstrate the robustness of the experimental procedure and suggest that the bending-based optical fringe method is highly effective for determining Young's modulus.

4.7 Summary of Findings

In summary, although the Poisson's ratio result shows a larger deviation from the expected value, the Young's modulus measurement exhibits strong agreement with standard references. These outcomes validate the experimental design, the precision of fringe displacement measurements, and the reliability of the data analysis approach. Ultimately, the combination of interference fringe tracking and beam-bending theory has proven to be a suitable and accurate technique for extracting elastic properties of glass.

DISCUSSION

As the applied load on the glass beam increased from 1 kg to 6 kg, a distinct compression of the interference fringe patterns was observed in both horizontal and vertical orientations. This trend, evident in **??**, corresponds to an increase in the beam's curvature, leading to a denser fringe distribution. The physical basis for this lies in beam-bending theory: as load increases, the bending moment $M = mga$ also increases, resulting in greater deformation of the beam and a smaller radius of curvature R. Consequently, the optical path difference per fringe increases, compressing the fringes spatially and making them appear closer together.

From the data in **??**, it is clear that the fringe displacement differences X and Y, and more importantly their squared values X_n^2 and Y_n^2, increase consistently with increasing of fringe index n and decrease consistently with increasing of load. This trend aligns with the theoretical model described in **??**, where the squared fringe separations follow the relationships

$$X_n^2 = 4\lambda R_x n - c_x \quad \text{and} \quad Y_n^2 = 4\lambda R_y n - c_y,$$

with R_x and R_y being the horizontal and vertical curvatures of the beam, respectively.

As the applied load increases, the beam experiences greater bending, which leads to a smaller radius of curvature. Consequently, the slopes of the X_n^2 vs. n and Y_n^2 vs. n plots become gentler. This compressive trend in fringe spacing confirms the validity of the optical method and the predictable elastic response of the glass beam.

The experimental data demonstrate a clear, nearly linear relationship between the squared fringe separations and the fringe order, thereby validating the theoretical predictions of Cornu's method. The slopes obtained from linear regressions in **??** yield the radii of curvature R_x and R_y, which are then used to compute Poisson's ratio via $\sigma = R_x/R_y$, and—together with the beam's geometric parameters and applied load—Young's modulus E.

Minor deviations from perfect linearity were observed and can be attributed to practical factors inherent in the experiment. Imperfections in the flatness of the glass surfaces for both the beam and the reference plate may lead to variations in the local air-gap thickness, thereby affecting the fringe pattern. Additionally, slight misalignments in the optical setup, particularly in the orientation of the stage telescope and the 45° mirror, can induce systematic errors in fringe position measurements. Ambient vibrations, whether originating from building activity or other laboratory disturbances, further contribute to fluctuations, especially at higher fringe orders where small errors are amplified.

These measurement uncertainties propagate through the analysis, affecting the determination of the slopes and, subsequently, the calculated radii of curvature, Poisson's ratio, and Young's modulus. Despite these challenges, careful error propagation in the linear regression analysis produced final values of $\sigma = 0.40 \pm 0.02$ and $E = 73.81 \pm 1.86\,\mathrm{GPa}$, which are broadly consistent with typical literature values for soda-lime glass.

To further evaluate the accuracy of these results, a comparison was made against standard reference values: the literature value for Poisson's ratio is approximately $\sigma_{\mathrm{ref}} = 0.30$, while the accepted range for Young's modulus lies between 65 and 75 GPa. The discrepancy in Poisson's ratio was calculated to be approximately 33.33%, indicating a systematic overestimation. This deviation may stem from experimental asymmetries, such as slight misalignments in vertical fringe tracking or optical distortion affecting the fringe radius estimation in the vertical plane.

On the other hand, the experimentally determined Young's modulus shows significantly better agreement with accepted values. The percentage discrepancy from the lower bound of the expected range (65 GPa) is 10.48%, and from the upper bound (75 GPa) is only 4.25%. These small deviations affirm that the experimental setup and computational procedures used to determine Young's modulus are both precise and reliable. They also highlight that while the experimental estimation of Poisson's ratio is more sensitive to vertical fringe measurements, the determination of E from horizontal curvature is considerably more robust and less prone to systematic error.

To further improve the experiment, enhanced surface preparation and the use of higher-quality optical flats could reduce errors caused by surface irregularities. Moreover, employing rigid, vibration-isolated mounts for the optical components and incorporating automated fringe detection with digital image processing would likely minimize misalignments and human errors. Future investigations might explore varying load conditions or examine alternative materials with different elastic properties. Additionally, extending the method to include temperature-dependent studies could provide deeper insights into material behavior under varying environmental conditions. Another promising improvement would be the use of polarized light or wavelength-tunable laser sources, which can enhance fringe contrast and resolution, thereby improving the precision of fringe measurement. These improvements would not only refine the accuracy of the current method but also broaden its applicability in material characterization studies.To further improve the experiment, enhanced surface preparation and the use of higher-quality optical flats could reduce errors caused by surface irregularities. Moreover, employing rigid, vibration-isolated mounts for the optical components and incorporating automated fringe detection with digital image processing would likely minimize misalignments and human errors. Future investigations might explore varying load conditions or examine alternative materials with different elastic properties. Additionally, extending the method to include temperature-dependent studies could provide deeper insights into material behavior under varying environmental conditions. Another promising improvement would be the use of polarized or wavelength-tunable laser sources, which can enhance fringe contrast and resolution, thereby improving the precision of fringe measurement.

Building on recent literature, such as the comprehensive study of Poisson's ratios across a variety of glasses, ceramics, and crystals by Kojima *et al.* [3], future work could explore the applicability of Cornu's method to more complex or anisotropic materials. The cited work highlights material-specific Poisson's ratios and the sensitivity of measurement techniques to underlying structure and bonding characteristics. Investigating whether optical interferometry methods such as Cornu's remain accurate across different crystalline symmetries or amorphous structures would be valuable for broadening the method's utility in material characterization.

These improvements would not only refine the accuracy of the current method but also expand its applicability to a wider range of engineering and materials science contexts.

CONCLUSION

The experiment successfully determined the elastic properties of a glass beam using Cornu's method based on optical interferometry. The average Poisson's ratio was found to be $\sigma = 0.40 \pm 0.02$, while the average Young's modulus was calculated as $E = 73.81 \pm 1.86\,\text{GPa}$. Although the Poisson's ratio exhibits a notable discrepancy of approximately 33.33% from the typical literature value of 0.30, the measured Young's modulus falls well within the accepted range of 65–75 GPa, with a discrepancy of only 10.48% from the lower bound and 4.25% from the upper bound.

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APPENDICES

8.1 Dataset

```
mass,orientation,n,Dif
1,horizontal,4,37.41
1,horizontal,3,28.46
1,horizontal,2,19.58
1,horizontal,1,7.95
1,vertical,2,31.34
1,vertical,1,18.21
3,horizontal,6,25.34
3,horizontal,5,21.77
3,horizontal,4,18.73
3,horizontal,3,15.29
3,horizontal,2,11.11
3,horizontal,1,4.38
3,vertical,4,35.90
3,vertical,3,31.74
3,vertical,2,26.91
3,vertical,1,19.50
4,horizontal,6,21.16
4,horizontal,5,18.85
4,horizontal,4,16.67
4,horizontal,3,13.13
4,horizontal,2,9.54
4,horizontal,1,3.69
4,vertical,6,36.63
4,vertical,5,34.55
4,vertical,4,29.95
4,vertical,3,24.95
4,vertical,2,18.3
4,vertical,1,3.15
5,horizontal,6,18.65
5,horizontal,5,16.56
5,horizontal,4,14.10
5,horizontal,3,11.14
5,horizontal,2,7.32
5,horizontal,1,1.35
5,vertical,6,35.33
5,vertical,5,31.72
5,vertical,4,27.85
```

```
5,vertical,3,23.60
5,vertical,2,17.78
5,vertical,1,6.49
6,horizontal,6,17.94
6,horizontal,5,16.23
6,horizontal,4,13.97
6,horizontal,3,11.55
6,horizontal,2,8.68
6,horizontal,1,3.38
6,vertical,6,35.17
6,vertical,5,30.77
6,vertical,4,27.72
6,vertical,3,23.51
6,vertical,2,19.14
6,vertical,1,11.38
```

8.2 Complete table

```python
import pandas as pd
import numpy as np

# Load the dataset (replace this with your file path or load directly)
df = pd.read_csv("measurement_data_with_Xn.csv")  # <-- Replace with your
    actual file name

# Standard uncertainty for both measurements
delta_combined = np.sqrt(0.01**2 + 0.01**2)

# Compute Dif^2 and $\delta$Dif^2
df["Dif_squared"] = df["Dif"] ** 2
df["delta_Dif_squared"] = 2 * df["Dif"].abs() * delta_combined

# Round for neat output
df["Dif"] = df["Dif"].round(2)
df["Dif_squared"] = df["Dif_squared"].round(2)
df["delta_Dif_squared"] = df["delta_Dif_squared"].round(2)

# Save the full dataset with computed columns
df.to_csv("measurement_data_with_XY.csv", index=False)

# Sort for presentation
df = df.sort_values(by=["mass", "orientation", "n"], ascending=[True, True,
    False])
grouped = df.groupby(["mass", "orientation"])

# Print formatted output
print("Data with computed squared differences labeled X or Y based on
    orientation:\n")
for (mass, orientation), group in grouped:
    is_horizontal = orientation == "horizontal"
    label = "X" if is_horizontal else "Y"

    print(f"\nTable: {mass} kg - {orientation.upper()}")
    print("-" * 50)
    print(group[["n", "Dif", "Dif_squared", "delta_Dif_squared"]]
          .rename(columns={
              "Dif": f"{label} (mm)",
              "Dif_squared": f"{label}^2 (mm^2)",
```

```
            "delta_Dif_squared": f"$\delta${label}^2 (mm^2)"
        })
        .to_string(index=False))
```

8.3 Output

```
Data with computed squared differences labeled X or Y based on orientation:



Table: 1 kg - HORIZONTAL
--------------------------------------------------
 n  X (mm)  X^2 (mm^2)  $\delta$X^2 (mm^2)
 4   37.41   1399.51       1.06
 3   28.46    809.97       0.80
 2   19.58    383.38       0.55
 1    7.95     63.20       0.22

Table: 1 kg - VERTICAL
--------------------------------------------------
 n  Y (mm)  Y^2 (mm^2)  $\delta$Y^2 (mm^2)
 2   31.34     982.2       0.89
 1   18.21     331.6       0.52

Table: 3 kg - HORIZONTAL
--------------------------------------------------
 n  X (mm)  X^2 (mm^2)  $\delta$X^2 (mm^2)
 6   25.34    642.12       0.72
 5   21.77    473.93       0.62
 4   18.73    350.81       0.53
 3   15.29    233.78       0.43
 2   11.11    123.43       0.31
 1    4.38     19.18       0.12

Table: 3 kg - VERTICAL
--------------------------------------------------
 n  Y (mm)  Y^2 (mm^2)  $\delta$Y^2 (mm^2)
 4   35.90   1288.81       1.02
 3   31.74   1007.43       0.90
 2   26.91    724.15       0.76
 1   19.50    380.25       0.55

Table: 4 kg - HORIZONTAL
--------------------------------------------------
 n  X (mm)  X^2 (mm^2)  $\delta$X^2 (mm^2)
 6   21.16    447.75       0.60
 5   18.85    355.32       0.53
```

```
 4   16.67    277.89       0.47
 3   13.13    172.40       0.37
 2    9.54     91.01       0.27
 1    3.69     13.62       0.10

Table: 4 kg - VERTICAL
----------------------------------------------------
 n  Y (mm)  Y^2 (mm^2)  $\delta$Y^2 (mm^2)
 6   36.63   1341.76       1.04
 5   34.55   1193.70       0.98
 4   29.95    897.00       0.85
 3   24.95    622.50       0.71
 2   18.30    334.89       0.52
 1    3.15      9.92       0.09

Table: 5 kg - HORIZONTAL
----------------------------------------------------
 n  X (mm)  X^2 (mm^2)  $\delta$X^2 (mm^2)
 6   18.65    347.82       0.53
 5   16.56    274.23       0.47
 4   14.10    198.81       0.40
 3   11.14    124.10       0.32
 2    7.32     53.58       0.21
 1    1.35      1.82       0.04

Table: 5 kg - VERTICAL
----------------------------------------------------
 n  Y (mm)  Y^2 (mm^2)  $\delta$Y^2 (mm^2)
 6   35.33   1248.21       1.00
 5   31.72   1006.16       0.90
 4   27.85    775.62       0.79
 3   23.60    556.96       0.67
 2   17.78    316.13       0.50
 1    6.49     42.12       0.18

Table: 6 kg - HORIZONTAL
----------------------------------------------------
 n  X (mm)  X^2 (mm^2)  $\delta$X^2 (mm^2)
 6   17.94    321.84       0.51
 5   16.23    263.41       0.46
 4   13.97    195.16       0.40
 3   11.55    133.40       0.33
```

```
  2    8.68      75.34        0.25
  1    3.38      11.42        0.10

Table: 6 kg - VERTICAL
--------------------------------------------------
  n  Y (mm)  Y^2 (mm^2)  $\delta$Y^2 (mm^2)
  6   35.17   1236.93        0.99
  5   30.77    946.79        0.87
  4   27.72    768.40        0.78
  3   23.51    552.72        0.66
  2   19.14    366.34        0.54
  1   11.38    129.50        0.32
```

8.4 Graphs Plots

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
import os

# Load the dataset
df = pd.read_csv("measurement_data_with_XY.csv")

# Ensure output folder exists
os.makedirs("plots", exist_ok=True)

# Sort and group
df = df.sort_values(by=["mass", "orientation", "n"], ascending=[True, True, True])
grouped = df.groupby(["mass", "orientation"])

# Adjustable error bar width (line width)
errorbar_width = 2  # <-- you can change this value

for (mass, orientation), group in grouped:
    is_horizontal = orientation == "horizontal"

    # Set labels and titles
    variable = "X" if is_horizontal else "Y"
    square_label = f"{variable}^2"
    square_label_units = f"{square_label} (mm^2)"
    plot_title = f"{mass} kg - {square_label} vs n"

    # Data
    x = group["n"].values
    y = group["Dif_squared"].values
    y_err = group["delta_Dif_squared"].values

    # Linear regression
    slope, intercept, r_value, p_value, std_err = stats.linregress(x, y)
    y_fit = slope * x + intercept

    # Std error of intercept
    n_points = len(x)
```

```
    x_mean = np.mean(x)
    s_xx = np.sum((x - x_mean) ** 2)
    std_err_intercept = std_err * np.sqrt(np.sum(x**2) / (n_points * s_xx))

    # Start plotting
    plt.figure(figsize=(8, 6))

    # Plot with red error bars and blue markers
    plt.errorbar(x, y, yerr=y_err, fmt='o', markersize=6, capsize=5,
                 ecolor='red', elinewidth=errorbar_width,
                 markerfacecolor='blue', markeredgecolor='black',
                 label="Data with uncertainty")

    # Regression line
    plt.plot(x, y_fit, color="darkorange", linestyle="--", linewidth=2, label="Linear Fit")

    # Labels and layout
    plt.title(plot_title)
    plt.xlabel("n")
    plt.ylabel(square_label_units)
    plt.grid(True)
    plt.legend()

    # Regression summary table
    table_data = [
        ["Gradient", f"{slope:.4f}"],
        ["Std. Dev (Grad)", f"{std_err:.4f}"],
        ["Y-Intercept", f"{intercept:.4f}"],
        ["Std. Dev (Y-Int)", f"{std_err_intercept:.4f}"],
        ["Corr Coef (R)", f"{r_value:.4f}"]
    ]

    # Add table to bottom
    plt.table(
        cellText=table_data,
        colLabels=["Parameter", "Value"],
        loc='bottom',
        cellLoc='center',
        bbox=[0.0, -0.45, 1, 0.35]
    )

```

```
    plt.subplots_adjust(bottom=0.35)

    # Save and show
    filename = f"plots/{mass}kg_{variable}2_vs_n_with_errorbar.png"
    plt.savefig(filename, dpi=300, bbox_inches='tight')
    plt.show()

    print(f" Saved plot with red error bars: {filename}")
```

8.5 Poisson's Ratio

```python
import pandas as pd
import numpy as np
from scipy import stats

# Load dataset
df = pd.read_csv("measurement_data_with_XY.csv")

# Constants
wavelength_m = 589.3e-9  # lambda in meters

# Sort and group
df = df.sort_values(by=["mass", "orientation", "n"])
grouped = df.groupby(["mass", "orientation"])

# Linear regressions
regression_results = []
for (mass, orientation), group in grouped:
    x = group["n"].values
    y = group["Dif_squared"].values
    slope, intercept, r_value, p_value, std_err = stats.linregress(x, y)
    regression_results.append({
        "mass": mass,
        "orientation": orientation,
        "slope": slope,
        "std_err": std_err
    })

# Split horizontal/vertical slopes
results_df = pd.DataFrame(regression_results)
horizontal_df = results_df[results_df["orientation"] == "horizontal"].set_index("mass")
vertical_df = results_df[results_df["orientation"] == "vertical"].set_index("mass")

# Compute Poisson's Ratio and Radii
poisson_list = []

for mass in sorted(set(horizontal_df.index) & set(vertical_df.index)):
    mx_mm2 = horizontal_df.loc[mass, "slope"]
    dmx_mm2 = horizontal_df.loc[mass, "std_err"]
```

```
    my_mm2 = vertical_df.loc[mass, "slope"]
    dmy_mm2 = vertical_df.loc[mass, "std_err"]

    mx = mx_mm2 /1000000                # convert to meters
    dmx = dmx_mm2 / 1000000
    my = my_mm2 / 1000000
    dmy = dmy_mm2 / 1000000

    Rx = mx / (4 * wavelength_m)
    dRx = dmx / (4 * wavelength_m)
    Ry = my / (4 * wavelength_m)
    dRy = dmy / (4 * wavelength_m)

    sigma = Rx / Ry
    dsigma = sigma * np.sqrt((dRx / Rx) ** 2 + (dRy / Ry) ** 2)

    # Store with scientific formatting for large values
    poisson_list.append({
        "mass": f"{mass} kg",
        "m_x": f"{mx_mm2:.2f} mm^2",
        "$\delta$m_x": f"{dmx_mm2:.2f} mm^2",
        "m_y": f"{my_mm2:.2f} mm^2",
        "$\delta$m_y": f"{dmy_mm2:.2f} mm^2",
        "R_x": f"{Rx:.2e} m",
        "$\delta$R_x": f"{dRx:.2e} m",
        "R_y": f"{Ry:.2e} m",
        "$\delta$R_y": f"{dRy:.2e} m",
        "$sigma$": f"{sigma:.2f}",
        "$\delta$$sigma$": f"{dsigma:.2f}"
    })

# Convert to DataFrame
poisson_df = pd.DataFrame(poisson_list)

# Compute mean Poisson's ratio
sigmas = [float(row["$sigma$"]) for row in poisson_list]
sigmas_err = [float(row["$\delta$$sigma$"]) for row in poisson_list]
mean_sigma = np.mean(sigmas)
mean_sigma_error = (1 / len(sigmas)) * np.sqrt(np.sum(np.square(sigmas_err)))

# Print summary table
```

```python
print("\n Poisson's Ratio Summary Table (with SI units and scientific notation)
    \n")
header = (
    f"{'Mass':<10} {'m_x (mm^2)':>12} {'$\delta$m_x (mm^2)':>12} "
    f"{'m_y (mm^2)':>12} {'$\delta$m_y (mm^2)':>12} "
    f"{'R_x (m)':>12} {'$\delta$R_x (m)':>12} {'R_y (m)':>12} {'$\delta$R_y (m)
    ':>12} "
    f"{'$sigma$':>6} {'$\delta$$sigma$':>6}"
)
print(header)
print("-" * len(header))

for row in poisson_list:
    print(f"{row['mass']:<10}{row['m_x']:>12}{row['$\delta$m_x']:>12}"
          f"{row['m_y']:>12}{row['$\delta$m_y']:>12}"
          f"{row['R_x']:>12}{row['$\delta$R_x']:>12}"
          f"{row['R_y']:>12}{row['$\delta$R_y']:>12}"
          f"{row['$sigma$']:>6}{row['$\delta$$sigma$']:>6}")

print("-" * len(header))
print(f"Average Poisson's Ratio: {mean_sigma:.2f} $\pm$ {mean_sigma_error:.2f}"
    )
```

8.6 Output

```
Poisson's Ratio Summary Table (with SI units and scientific notation)

Mass           m_x (mm^2)   $\delta$m_x (mm^2)    m_y (mm^2)   $\delta$m_y (mm^2)      R_x (m)      $\delta$R_x (m)       R_y
    (m)      $\delta$R_y (m)       $sigma$      $\delta$$sigma$
-----------------------------------------------------------------------------------------------------------------------------
1 kg         443.55 mm^2   42.78 mm^2  650.60 mm^2    0.00 mm^2  1.88e+02 m  1.81e+01 m  2.76e+02 m  0.00e+00 m  0.68  0.07
3 kg         122.38 mm^2    5.21 mm^2  300.90 mm^2   10.72 mm^2  5.19e+01 m  2.21e+00 m  1.28e+02 m  4.55e+00 m  0.41  0.02
4 kg          87.69 mm^2    1.81 mm^2  271.72 mm^2   12.67 mm^2  3.72e+01 m  7.68e-01 m  1.15e+02 m  5.38e+00 m  0.32  0.02
5 kg          70.48 mm^2    1.98 mm^2  237.69 mm^2    3.94 mm^2  2.99e+01 m  8.40e-01 m  1.01e+02 m  1.67e+00 m  0.30  0.01
6 kg          62.23 mm^2    0.62 mm^2  214.12 mm^2    6.99 mm^2  2.64e+01 m  2.65e-01 m  9.08e+01 m  2.97e+00 m  0.29  0.01
-----------------------------------------------------------------------------------------------------------------------------
Average Poisson's Ratio: 0.40 $\pm$ 0.02
```

8.7 Young's Modulus

```
import pandas as pd
import numpy as np
from scipy import stats

# Load dataset
df = pd.read_csv("measurement_data_with_XY.csv")

# Experiment constants (user measured, converted to SI)
a = 4.4575 / 100       # cm -> m
da = 0.005 / 100
b = 5.115 / 100        # cm -> m
db = 0.005 / 100
d = 6.06 / 1000        # mm -> m
dd = 0.01 / 1000
g = 9.81               # m/s^2
$\lambda$ = 589.3e-9          # m

# Sort and group data
df = df.sort_values(by=["mass", "orientation", "n"])
grouped = df.groupby(["mass", "orientation"])

# Regression to extract slope
regression_results = []
for (mass, orientation), group in grouped:
    x = group["n"].values
    y = group["Dif_squared"].values
    slope, _, _, _, std_err = stats.linregress(x, y)
    regression_results.append({
        "mass": mass,
        "orientation": orientation,
        "slope": slope,
        "std_err": std_err
    })

# Filter only horizontal data for Young's modulus
results_df = pd.DataFrame(regression_results)
horizontal_df = results_df[results_df["orientation"] == "horizontal"].set_index
    ("mass")

# Compute Young's modulus for each mass
```

```python
young_list = []
for mass in sorted(horizontal_df.index):
    m = mass
    mx_mm2 = horizontal_df.loc[m, "slope"]      # mm^2
    dmx_mm2 = horizontal_df.loc[m, "std_err"]   # mm^2

    mx = mx_mm2 /1000000                  # convert to meters
    dmx = dmx_mm2 / 1000000

    Rx = mx / (4 * $\lambda$)                        # R_x in m
    dRx = dmx / (4 * $\lambda$)

    E = (12 * m * g * a * Rx) / (b * d**3)           # Young's modulus (Pa)
    dE = E * np.sqrt(
        (dRx / Rx)**2 +
        (da / a)**2 +
        (db / b)**2 +
        (3 * dd / d)**2
    )

    E_GPa = E / 1e9
    dE_GPa = dE / 1e9

    young_list.append({
        "mass": f"{m} kg",
        "m_x": f"{mx_mm2:.2f} mm^2",
        "$\delta$m_x": f"{dmx_mm2 :.2f} mm^2",
        "R_x": f"{Rx*1000:.2e} mm",
        "$\delta$R_x": f"{dRx*1000:.2e} mm",
        "E": f"{E_GPa:.3f} GPa",
        "$\delta$E": f"{dE_GPa:.3f} GPa"
    })

# Convert to DataFrame for final display
young_df = pd.DataFrame(young_list)

# Compute mean and uncertainty
E_mean = np.mean([float(e["E"].split()[0]) for e in young_list])
dE_mean = (1 / len(young_list)) * np.sqrt(np.sum([
    float(e["$\delta$E"].split()[0])**2 for e in young_list
]))

```

```
# Print formatted output
print("\n Young's Modulus Summary Table (with SI units)\n")
header = f"{'Mass':<10}{'m_x':>12}{'$\delta$m_x':>12}{'R_x':>14}{'$\delta$R_x
    ':>14}{'E':>14}{'$\delta$E':>14}"
print(header)
print("-" * len(header))

for row in young_list:
    print(f"{row['mass']:<10}{row['m_x']:>12}{row['$\delta$m_x']:>12}"
          f"{row['R_x']:>14}{row['$\delta$R_x']:>14}{row['E']:>14}{row['$\
    delta$E']:>14}")

print("-" * len(header))
print(f"Average Young's Modulus: {E_mean:.3f} $\pm$ {dE_mean:.3f} GPa")
```

8.8 Output

```
Young's Modulus Summary Table (with SI units)

Mass                   m_x          $\delta$m_x            R_x           $\delta$R_x              E             $\delta$E
------------------------------------------------------------------------------------------
1 kg         443.55 mm^2    42.78 mm^2    1.88e+05 mm    1.81e+04 mm     86.741 GPa      8.377 GPa
3 kg         122.38 mm^2     5.21 mm^2    5.19e+04 mm    2.21e+03 mm     71.797 GPa      3.081 GPa
4 kg          87.69 mm^2     1.81 mm^2    3.72e+04 mm    7.68e+02 mm     68.593 GPa      1.459 GPa
5 kg          70.48 mm^2     1.98 mm^2    2.99e+04 mm    8.40e+02 mm     68.912 GPa      1.968 GPa
6 kg          62.23 mm^2     0.62 mm^2    2.64e+04 mm    2.65e+02 mm     73.019 GPa      0.825 GPa
------------------------------------------------------------------------------------------
Average Young's Modulus: 73.812 $\pm$ 1.858 GPa
```

8.9 Discrepancy Calculation

```python
def percentage_discrepancy(measured, reference):
    """
    Calculate percentage discrepancy from a single reference value.
    """
    return abs((measured - reference) / reference) * 100

def discrepancy_from_range(measured, lower, upper):
    """
    Calculate percentage discrepancies from both ends of a reference range.
    """
    lower_discrepancy = abs((measured - lower) / lower) * 100
    upper_discrepancy = abs((measured - upper) / upper) * 100
    return lower_discrepancy, upper_discrepancy

# === Input your data here ===
# Example values
measured_poisson = 0.40
reference_poisson = 0.30

measured_E = 71.81  # GPa (your measured Young's modulus)
reference_E_range = (65, 75)  # GPa (typical range for glass)

# === Calculation ===
# Poisson Ratio
poisson_discrepancy = percentage_discrepancy(measured_poisson, reference_poisson)
print(f"Poisson's Ratio Discrepancy: {poisson_discrepancy:.2f}%")

# Young's Modulus
E_low, E_high = reference_E_range
E_disc_low, E_disc_high = discrepancy_from_range(measured_E, E_low, E_high)
print(f"Young's Modulus Discrepancy from Lower Bound ({E_low} GPa): {E_disc_low:.2f}%")
print(f"Young's Modulus Discrepancy from Upper Bound ({E_high} GPa): {E_disc_high:.2f}%")
```

8.10 Output

```
Poisson's Ratio Discrepancy: 33.33%
Young's Modulus Discrepancy from Lower Bound (65 GPa): 10.48%
Young's Modulus Discrepancy from Upper Bound (75 GPa): 4.25%
```